

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Summer 2023

Course: B. Tech.Branch:Civil Engineering

Semester :5th Sem

Subject Code & Name: BTCVC504_Concrete technology

Max Marks: 60

Date:17/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/ Marks
CO)

Q. 1 Solve Any Two of the following.

- | | | | |
|-----------|--|------------|----------|
| A) | List the properties of fine aggregate and explain one of them in detail. | CO1 | 6 |
| B) | Define bulking of sand and explain in detail the procedure to find the bulking of sand with diagram. | CO1 | 6 |
| C) | Explain the procedure to determine standard consistency of cement. | CO1 | 6 |

Q.2 Solve Any Two of the following.

- | | | | |
|-----------|---|------------|----------|
| A) | Describe the procedure for determination of workability by compaction factor test with diagram. | CO1 | 6 |
| B) | Explain the procedure of conducting a rebound hammer test for measuring the surface hardness with reference to following points : (i) Rebound number and its significance (ii) Construction of equipment (iii) Sketch of equipment (iv) limitations of test | CO1 | 6 |
| C) | Define the following terms :a) Setting time of cement b) Bleeding c) Workability
d) Durability e)Compressive Strength f) Fineness of cement | CO1 | 6 |

Q. 3 Solve Any Two of the following.

- | | | | |
|-----------|--|------------|----------|
| A) | Define Admixtures. Enlist its types and state purpose of adding the admixtures into the concrete. (any four) | CO2 | 6 |
| B) | Define concrete mix design. Write any four factors affecting on it and state any two methods of concrete mix design. State necessity of concrete mix design. | CO3 | 6 |
| C) | Calculate fineness modulus for the given data of fine aggregate. Total weight of C.A. = 1000 gm. State type of sand. | CO1 | 6 |

Sieve size (mm)	4.75	2.36	1.18	600 μ	300 μ	150 μ	Pan			
Weight retained (gm.)	100	150	300	200	120	90	40	75	210	27

Q.4 Solve Any Two of the following.

- A)** Explain Duff Abraham's water cements ratio law. **CO 3 6**
- B)** Define batching with its type. Draw concreting operation chain in sequence & Calculate quantity of water to be added for costing cubes in laboratory by 10 kg cement, if w/c ratio 0.5. **CO 3 6**
- C)** Explain the procedure to determine the compressive strength of concrete. **CO1 6**

Q. 5 Solve Any one of the following.

- A)** a) Explain the method of concrete mix design procedure by I.S. method. **CO3 12**
b) Illustrate the effect of properties of coarse aggregates on compressive strength of concrete.
- B)** Design a concrete mix for M45 grade using following data: **CO3 12**
a) Grade designation : M45
b) Type of cement : OPC 43 Grade (Specific Gravity=3.15)
c) Maximum nominal size of aggregate : 20mm (Specific Gravity=2.80)
d) Minimum cement content : 320 kg/m³
e) Maximum water-cement ratio : 0.45
f) Workability : 125 mm slump
g) Exposure condition : Severe (RCC)
h) Method of concrete placing : Pumping
i) Degree of supervision : Good
j) Type of aggregate : Crushed Angular Aggregates
k) Super Plasticizer will be used.
l) Fine aggregates: Confirming to Zone II (Specific Gravity=2.70)

Table 1 Value of X
(Clause 4.2)

Sl No.	Grade of Concrete	Value of X
(1)	(2)	(3)
i)	M10 M15	5.0
ii)	M20 M25	5.5
iii)	M30 M35 M40 M45 M50 M55 M60	6.5
iv)	M65 and above	8.0

Table 2 Assumed Standard Deviation
(Clause 4.2.1.3)

Sl No.	Grade of Concrete	Assumed Standard Deviation N/mm ²
(1)	(2)	(3)
i)	M10 M15	3.5
ii)	M20 M25	4.0
iii)	M30 M35 M40 M45 M50 M55 M60	5.0
iv)	M65 M70 M75 M80	6.0

Table 3 Approximate Air Content
(Clause 5.2)

Sl No.	Nominal Maximum Size of Aggregate mm	Entrapped Air, as Percentage of Volume of Concrete
(1)	(2)	(3)
i)	10	1.5
ii)	20	1.0
iii)	40	0.8

Table 4 Water Content per Cubic Metre of Concrete For Nominal Maximum Size of Aggregate
(Clause 5.3)

Sl No.	Nominal Maximum Size of Aggregate mm	Water Content ¹⁾ kg
(1)	(2)	(3)
i)	10	208
ii)	20	186
iii)	40	165

¹⁾Water content corresponding to saturated surface dry aggregate.

NOTES

1 These quantities of mixing water are for use in computing cement/cementitious materials content for trial batches.

2 On account of long distances over which concrete needs to be carried from batching plant/RMC plant, the concrete mix is generally designed for a higher slump initially than the slump required at the time of placing. The initial slump value shall depend on the distance of transport and loss of slump with time.

Table 5 Volume of Coarse Aggregate per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate for Water-Cement/Water-Cementitious Materials Ratio of 0.50
(Clause 5.5)

Sl No.	Nominal Maximum Size of Aggregate mm	Volume of Coarse Aggregate per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate			
		Zone IV	Zone III	Zone II	Zone I
(1)	(2)	(3)	(4)	(5)	(6)
i)	10	0.54	0.52	0.50	0.48
ii)	20	0.66	0.64	0.62	0.60
iii)	40	0.73	0.72	0.71	0.69

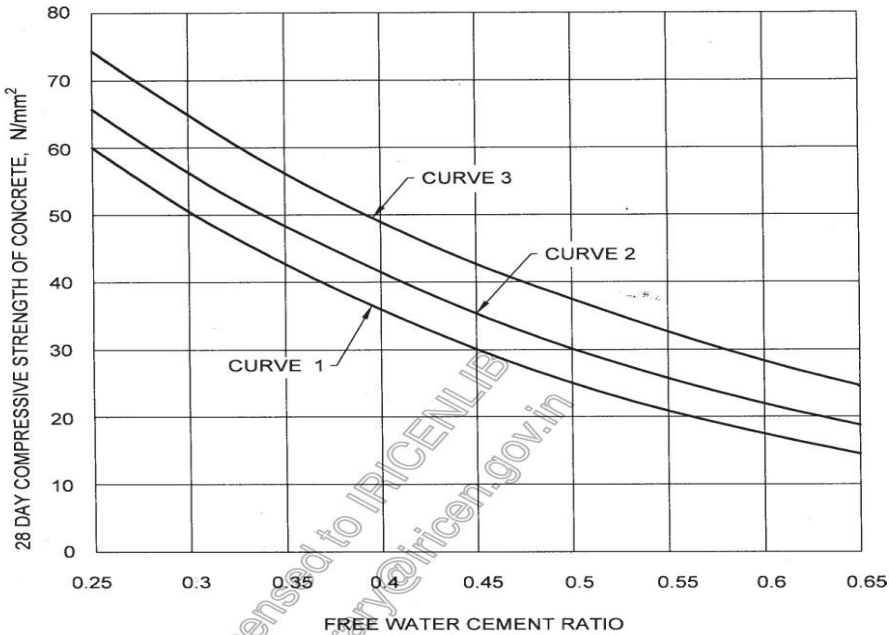
NOTES

1 Volumes are based on aggregates in saturated surface dry condition.

2 These volumes are for crushed (angular) aggregate and suitable adjustments may be made for other shape of aggregate.

3 Suitable adjustments may also be made for fine aggregate from other than natural sources, normally, crushed sand or mixed sand may need lesser fine aggregate content. In that case, the coarse aggregate volume shall be suitably increased.

4 It is recommended that fine aggregate conforming to Grading Zone IV, as per IS 383 shall not be used in reinforced concrete unless tests have been made to ascertain the suitability of proposed mix proportions.



Curve 1 : for expected 28 days compressive strength of 33 and < 43 N/mm².
Curve 2 : for expected 28 days compressive strength of 43 and < 53 N/mm².
Curve 3 : for expected 28 days compressive strength of 53 N/mm² and above.

*** End ***

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